

# Selection of Feature-Sparse Ensembles: Co-Error, Certification, and Generalization

Research draft

Living development manuscript; not a submission-ready evaluation

## Abstract

We investigate ensembles selected from large populations of feature-sparse weak predictors. The study asks whether empirical skill screening and explicit error complementarity improve generalization beyond random subspaces and ordinary validation-based ensemble selection. Squared-error and binary Brier losses admit an exact residual second-moment objective, whereas correlation-based diversity discards scale and can penalize useful negative dependence. The protocol reserves outer test folds, compares methods on shared candidate populations, and records selection and computational costs. An initial synthetic study finds modest gains over quality selection, severe screening infeasibility, and false survivors under the null, while strong forest and boosting baselines generally perform better. An oracle additive counterexample identifies attenuation from averaging subspaces. A follow-up affine diagnostic helps fixed subsets, but profiled selection improves OOF loss while usually worsening test loss. Candidate-pool scaling reveals conditional gains and increased selection optimism. Results remain development evidence; no benchmark superiority or novelty claim is established.

## 1 Introduction

Useful tabular information may be distributed across weakly informative subspaces. Selecting complementary errors could improve an average of weak predictors, but this hypothesis competes with simpler explanations: regularization, more search compute, and ordinary ensemble selection. We test these alternatives using paired candidate pools and known synthetic mechanisms. Sparse signal and interactions provide explicit potential counterexamples.

## 2 Related Work and Unresolved Novelty

Bagging [Breiman, 1996a] and random forests [Breiman, 2001] reduce instability through aggregation. Random subspaces [Ho, 1998] assign fixed feature subsets to trees; random patches [Louppe and Geurts, 2012] also sample rows. RaSE [Tian and Feng, 2021] already generates and selects random subspaces, making a claim to novelty based on subspace search alone untenable. Ensemble selection [Caruana et al., 2004] optimizes validation ensemble performance from a model library. The ambiguity decomposition [Krogh and Vedelsby, 1995] motivates squared-loss aggregation; diversity can help or hurt majority votes [Brown and Kuncheva, 2010]. These precedents require differentiating any eventual contribution through mechanism, statistical procedure, scaling, or a substantiated algorithmic improvement. Quadratic pruning with cardinality constraints and SDP relaxations predates this study [Zhang et al., 2006]; the optimization formulation alone is not a contribution. Stacked regressions [Breiman, 1996b] fit nonnegative coefficients on cross-validation predictions and explicitly formulate simplex weighting using residual-product quadratic loss. Our co-error matrix and scalar affine diagnostic therefore have close stacking precedents; restricting weights to a selected, equal-weight subset changes the constraint, not the loss identity. Portfolio weighting explicitly uses prediction-error covariances [Pérez-Rodríguez et al., 2023]. Stacking shrinkage theory for honest, nested least-squares subspaces [Chen et al., 2023] provides a useful regularization connection, but its assumptions do not cover our non-nested fitted sparse trees. Negative correlation learning jointly trains interacting

networks [Liu and Yao, 1999], whereas our current models are generated independently and selected afterward. Random-projection ensembles also combine randomized search, validation selection, and aggregation [Cannings and Samworth, 2017]. Diversity penalties can change effective capacity [Reeve and Brown, 2018], motivating a regularization explanation. Cross-validation uncertainty is subtle under overlapping fits [Bengio and Grandvalet, 2004, Bates et al., 2024]. Portfolio-based weighting, conditional dependence, and selective inference require further full-text review.

### 3 Method

Candidate  $b$  has a fixed raw-feature mask  $F_b$ , tree depth, leaf-size constraint, row-sampling fraction, and seed, specified before fitting. Fold-local preprocessing and the tree see only  $F_b$ . Inner cross-fitting yields predictions  $P_{nb}$ . Binary screening uses a stratified empirical bootstrap lower AUROC quantile; regression uses cross-fitted null-relative squared-error skill. We retain this as a heuristic screen, not a familywise certificate.

#### 3.1 Theoretical Motivation

Let  $E_{nb} = y_n - P_{nb}$  and  $G = E^\top E/N$ . For an equal-weight subset  $S$  of cardinality  $K$ ,

$$\hat{L}_{\text{sq}}(S) = K^{-2} \sum_{i,j \in S} G_{ij}. \quad (1)$$

This is an algebraic identity for regression MSE and binary Brier loss, not a generalization bound. Writing  $\mu = N^{-1}E^\top \mathbf{1}$  gives  $G = C + \mu\mu^\top$ . Correlation removes marginal scale and mean; covariance alone removes bias. Neither equals squared ensemble loss. Statistical independence does not follow from zero correlation. Prediction correlation, error correlation, and useful ensemble complementarity are distinct.

We compare random selection, empirical quality, absolute and signed error correlation, minimax dependence, quality–dependence tradeoffs, and greedy co-error followed by improving one-swaps. Direct squared-loss selection without replacement is exactly the same greedy objective, so it supplies a verification baseline rather than a separate contribution. A with-replacement Caruana-style baseline yields nonuniform weights and is labeled separately.

#### 3.2 Bias-Preserving Shrinkage Variant

We additionally test  $G_\alpha = (1 - \alpha)C + \alpha \text{diag}(C) + \mu\mu^\top$ . Shrinkage could reduce noisy cross-moment exploitation as the pool grows. It can also suppress genuine complementarity. The falsifying experiment holds  $K$  fixed, varies  $B/N$ , and measures outer loss and selection optimism. This regularizer has no established novelty or superiority claim.

#### 3.3 An Oracle Attenuation Counterexample

Suppose  $Y = \sum_{j=1}^d s_j(X_j) + \epsilon$ , with independent centered signal components of equal variance  $v$ , independent centered noise of variance  $\sigma^2$ , and population-optimal subspace learners  $f_b = \mathbb{E}[Y \mid X_{F_b}] = \sum_{j \in F_b} s_j(X_j)$ . Define  $w_j = K^{-1} \sum_{b \in S} \mathbb{I}(j \in F_b)$ . Then

$$L(S) = \sigma^2 + v \sum_{j=1}^d (1 - w_j)^2. \quad (2)$$

If each mask contains at most  $q < d$  informative features, then  $\sum_j w_j \leq q$  and Jensen’s inequality yields  $L(S) \geq \sigma^2 + dv(1 - q/d)^2$ . Thus averaging individually optimal subspace predictors attenuates distributed additive signal, even when their union covers all informative coordinates and  $B, K$  are arbitrarily large. For the additive regression development design,  $d = 16$ ,  $v = 1/16$ ,  $q \leq 4$ , and  $\sigma^2 = 1$ , so this idealized conditional-mean ensemble has risk at least 1.5625. This is a population calculation for these oracle learners,

not a lower bound for arbitrary subspace functions, finite-sample fitted trees, or classification AUROC. Coordinated fitting or non-simplex rescaling can escape its assumptions. It motivates testing aggregate scale calibration separately from diversity.

### 3.4 Affine-Profiled Diagnostic Variant (APCE)

After Experiment 001, we test an affine correction  $a\bar{P}_S + b$ , with  $a \geq 0$  and intercept  $b$  fitted exclusively on training OOF predictions. For a fixed subset, the profiled squared loss is

$$\widehat{L}_{\text{aff}}(S) = \widehat{\text{Var}}(Y) - \frac{\max(\widehat{\text{Cov}}(Y, \bar{P}_S), 0)^2}{\widehat{\text{Var}}(\bar{P}_S)}. \quad (3)$$

Zero aggregate variance gives the constant predictor. Centered prediction moments support incremental greedy and one-swap updates. The comparison includes affine correction of unchanged random, top-quality, and co-error subsets, isolating selection from the two new aggregation parameters. This is a stacking-like diagnostic and is not claimed as a novel optimizer or equal-output-weight primary method.

## 4 Experimental Protocol

All initial settings are fixed before each run. Outer folds remain unused during candidate screening and selection; selected specifications are refitted on the full outer training fold. Any future tuning regenerates the complete pipeline inside an additional tuning split, rather than slicing a globally fitted OOF matrix. All baselines share outer splits and fold-local preprocessing. Development synthetic regimes are separate from any locked real-world confirmation suite. Binary AUROC and regression RMSE are primary; multiclass log loss is primary. Brier loss is additionally central to the mechanism analysis. Bootstrap intervals over fixed OOF rows omit training instability, and unadjusted candidate screens omit multiplicity. These limitations prohibit interpreting them as simultaneous statistical certification. Shared candidate-search cost is charged to every standalone selector. Capacity and inference measurements accompany accuracy; matched-budget claims require dedicated experiments. Paired task-level analyses average folds and seeds before resampling tasks. No row-level significance claims across tasks are permitted.

## 5 Synthetic Experiments

Experiment 001 evaluated nine known mechanisms for each of binary classification and regression: additive, redundant, correlated blocks, sparse, dominant, XOR, four-way parity, mixed, and null. Fixed settings were  $N = 600$ ,  $M = 40$ ,  $B = 100$ ,  $K = 8$ , feature fraction 0.1, tree depth 3, minimum leaf size 5, two generation seeds, three outer folds, and three OOF folds. This produced 108 paired outer evaluations. Forest and boosting baselines use static configurations; search compute, final capacity, and inference budgets are measured but unmatched.

The initial hypothesis is not supported as a broad performance claim at these settings. Unfiltered co-error improves mean binary AUROC over unfiltered top quality by 0.0107, but generally trails the strong forests and boosting models. Co-error does not consistently improve over absolute residual correlation on outer AUROC or squared loss; a positive aggregate Brier effect is concentrated in the dominant-feature regime. The  $\alpha = 0.5$  shrinkage variant fails to improve on unregularized co-error in the compared task means. These are negative findings for the preregistered settings, not universal statements about the objectives.

### 5.1 Screening Failure and Feasibility

Screened co-error is feasible in only 40 of 54 binary splits and 16 of 54 regression splits. On every binary null split, 3–13 of 100 candidates pass the empirical screen, despite population AUROC 0.5 under this data-generating process. The bootstrap lower quantile must therefore remain labeled an empirical screen. Across shared feasible splits, top-quality screening has almost no incremental benefit over unscreened top quality; it frequently prevents a fixed-size ensemble.

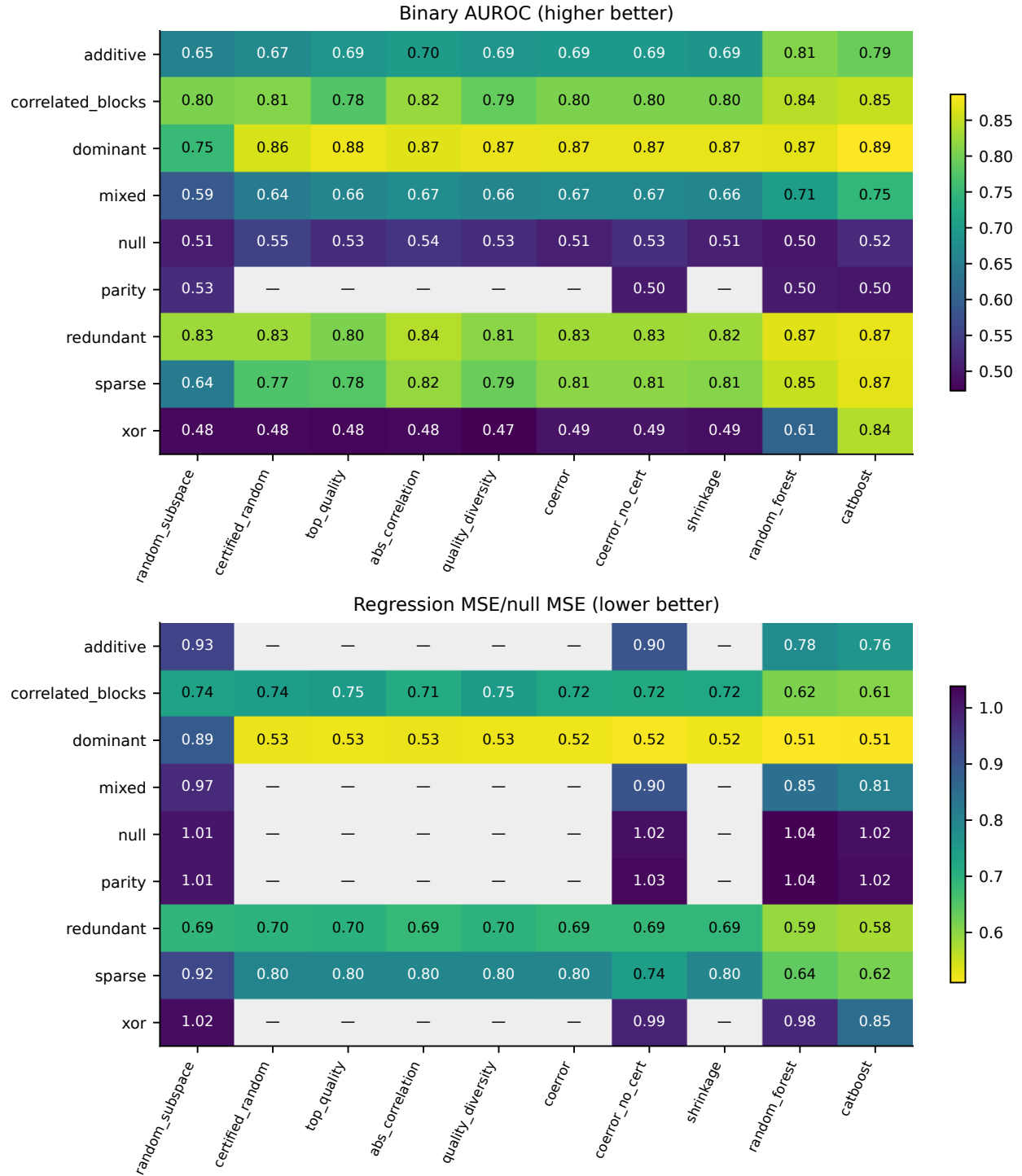


Figure 1: Development mechanism results. Gray cells indicate no feasible fixed- $K$  screened ensemble. Numerical annotations are means of completed splits; the tables mark partial feasibility. Rankings are not confirmation evidence.

| Regime            | RS    | Top                | AbsCorr            | CoErr              | CoErr-all | RF    | CatBoost |
|-------------------|-------|--------------------|--------------------|--------------------|-----------|-------|----------|
| additive          | 0.655 | 0.689              | 0.701              | 0.693              | 0.693     | 0.809 | 0.794    |
| correlated_blocks | 0.805 | 0.775              | 0.822              | 0.804              | 0.804     | 0.842 | 0.849    |
| dominant          | 0.752 | 0.875 <sup>†</sup> | 0.867 <sup>†</sup> | 0.872 <sup>†</sup> | 0.872     | 0.869 | 0.886    |
| mixed             | 0.588 | 0.662              | 0.672              | 0.666              | 0.666     | 0.705 | 0.746    |
| null              | 0.506 | 0.527 <sup>†</sup> | 0.535 <sup>†</sup> | 0.509 <sup>†</sup> | 0.529     | 0.500 | 0.522    |
| parity            | 0.528 | –                  | –                  | –                  | 0.503     | 0.502 | 0.500    |
| redundant         | 0.826 | 0.805              | 0.840              | 0.827              | 0.827     | 0.868 | 0.872    |
| sparse            | 0.645 | 0.778              | 0.820              | 0.815              | 0.815     | 0.849 | 0.868    |
| xor               | 0.483 | 0.478 <sup>†</sup> | 0.481 <sup>†</sup> | 0.494 <sup>†</sup> | 0.495     | 0.605 | 0.838    |

Table 1: exp\_001\_mechanisms\_v1: binary, auroc. Means over completed outer folds and seeds. –: fixed-K infeasible in every split; †: partial feasibility. Full coverage is stored separately. Static baselines, unmatched compute; development only.

| Regime            | RS    | Top                | AbsCorr            | CoErr              | CoErr-all | RF    | CatBoost |
|-------------------|-------|--------------------|--------------------|--------------------|-----------|-------|----------|
| additive          | 0.933 | –                  | –                  | –                  | 0.899     | 0.784 | 0.764    |
| correlated_blocks | 0.735 | 0.750 <sup>†</sup> | 0.713 <sup>†</sup> | 0.723 <sup>†</sup> | 0.718     | 0.615 | 0.606    |
| dominant          | 0.890 | 0.525 <sup>†</sup> | 0.525 <sup>†</sup> | 0.523 <sup>†</sup> | 0.515     | 0.511 | 0.514    |
| mixed             | 0.974 | –                  | –                  | –                  | 0.904     | 0.846 | 0.814    |
| null              | 1.007 | –                  | –                  | –                  | 1.018     | 1.038 | 1.018    |
| parity            | 1.009 | –                  | –                  | –                  | 1.027     | 1.039 | 1.023    |
| redundant         | 0.691 | 0.701              | 0.686              | 0.687              | 0.687     | 0.591 | 0.580    |
| sparse            | 0.925 | 0.797 <sup>†</sup> | 0.795 <sup>†</sup> | 0.797 <sup>†</sup> | 0.744     | 0.639 | 0.617    |
| xor               | 1.021 | –                  | –                  | –                  | 0.987     | 0.978 | 0.852    |

Table 2: exp\_001\_mechanisms\_v1: regression, normalized\_squared\_loss. Means over completed outer folds and seeds. –: fixed-K infeasible in every split; †: partial feasibility. Full coverage is stored separately. Static baselines, unmatched compute; development only.

## 5.2 Experiment 002: Attenuation Versus Selection

The follow-up fixes the candidate pools, masks, seeds, splits, and  $B/K$  settings from Experiment 001. It compares raw versus OOF-affine-calibrated random, top-quality, and co-error subsets, plus APCE, on six regression mechanisms. All subsets are unscreened to remove infeasibility as a confound. Each affine control selects exactly the same learners as its raw counterpart. No aggregation hyperparameter is tuned; these are two learned scalar parameters.

Affine calibration of co-error improves task-mean normalized loss in 5 of six regimes. Additive loss changes from 0.899 to 0.878, and mixed loss from 0.904 to 0.892. These improvements leave a large gap to the unrestricted forest on additive signal (0.784). The null control worsens from 1.018 to 1.031. Calibration therefore addresses part of the weakness without establishing a broadly competitive method or a unique attenuation explanation.

APCE improves or ties calibrated OOF loss relative to affine co-error in all 36 splits, yet loses on outer normalized loss in 5 of six task means and wins only the redundant regime. On additive signal its outer loss is 0.903, versus 0.878 for calibration of the unchanged co-error subset. The additional optimization improves the reused training criterion while usually hurting generalization. This is consistent with selection optimism; smaller training fits for OOF predictions and the subsequent refit remain alternatives.

## 5.3 Experiment 003: Candidate Search at Fixed Ensemble Size

We vary  $B \in \{30, 100, 300\}$  at fixed  $K = 8$  on additive and null mechanisms for both tasks, holding all other settings fixed. Across the 48 adjacent-pool fold comparisons, masks, OOF predictions, and bootstrap columns agree exactly on their prefixes; the fixed forest also gives identical predictions. Larger pools improve binary additive co-error performance, while additive regression is non-monotone: normalized co-error loss is

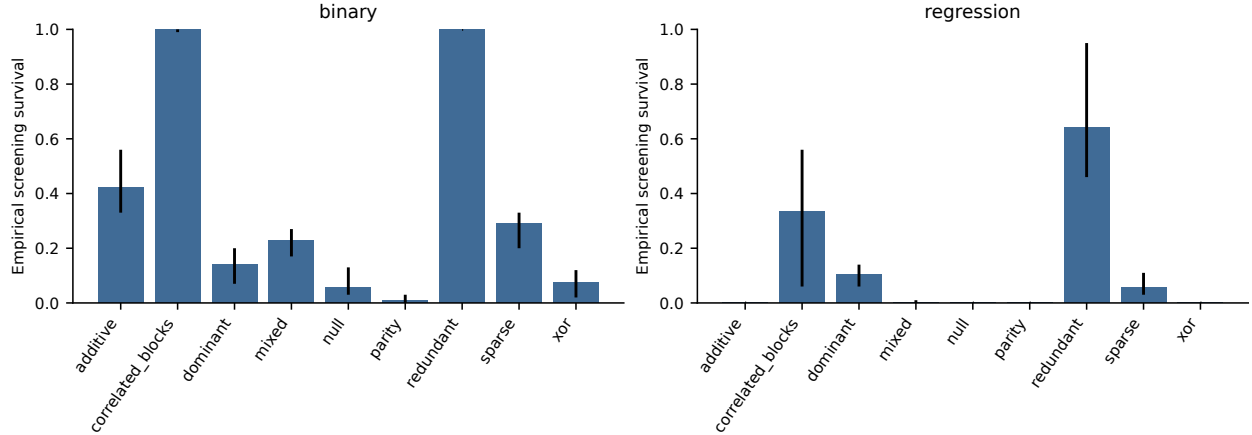


Figure 2: Empirical screening survival by regime. Error bars give the observed minimum and maximum over folds/seeds, not confidence intervals.

| Regime    | RS    | RS+aff | Top   | Top+aff | CoErr | CoErr+aff | APCE  |
|-----------|-------|--------|-------|---------|-------|-----------|-------|
| additive  | 0.933 | 0.922  | 0.920 | 0.912   | 0.899 | 0.878     | 0.903 |
| dominant  | 0.890 | 0.818  | 0.517 | 0.512   | 0.515 | 0.510     | 0.520 |
| mixed     | 0.974 | 0.984  | 0.937 | 0.929   | 0.904 | 0.892     | 0.902 |
| null      | 1.007 | 1.000  | 1.014 | 1.024   | 1.018 | 1.031     | 1.038 |
| redundant | 0.691 | 0.648  | 0.701 | 0.687   | 0.687 | 0.655     | 0.641 |
| sparse    | 0.925 | 0.889  | 0.775 | 0.749   | 0.744 | 0.673     | 0.685 |

Table 3: Experiment 002: regression MSE divided by training-mean null MSE, averaged over six outer evaluations per regime. All subsets are unscreened. +aff fits two aggregate parameters on training OOF predictions; APCE also changes selection.  $B=100$  and  $K=8$  are fixed. Lower is better. Development evidence only.

0.908, 0.899, and 0.903, respectively.

The flexible profiled selector exposes increasing optimism. Its additive training OOF MSE decreases from 1.602 through 1.499 to 1.396, while outer normalized loss changes from 0.864 to 0.903 and 0.896. The endpoint deterioration occurs for both generation seeds in additive and null regression. Fixed  $\alpha = 0.5$  shrinkage supplies no consistent outer remedy. Thus a larger library improves the reused criterion without a monotone generalization benefit for these selectors.

Binary null survivors rise from 0–5 at  $B = 30$  to 7–39 at  $B = 300$ . Regression null screening retains no candidates at any pool size. This separates multiplicity-related false survival from the regression screen’s low feasibility/power. It does not estimate familywise coverage from six splits. All comparisons reuse development seeds and remain descriptive.

## 6 Real-World Benchmarks

No confirmation benchmark has been evaluated. Current benchmark versions and baseline requirements will be recorded before locking the study.

## 7 Ablations and Analysis

The stored ablations include all required v0 components, signed versus absolute residual correlation, prediction correlation, covariance without bias, direct squared-loss greedy selection, with-replacement selection, and shrinkage. An artifact audit reconstructed loss and test metrics for all selected methods; the maxi-

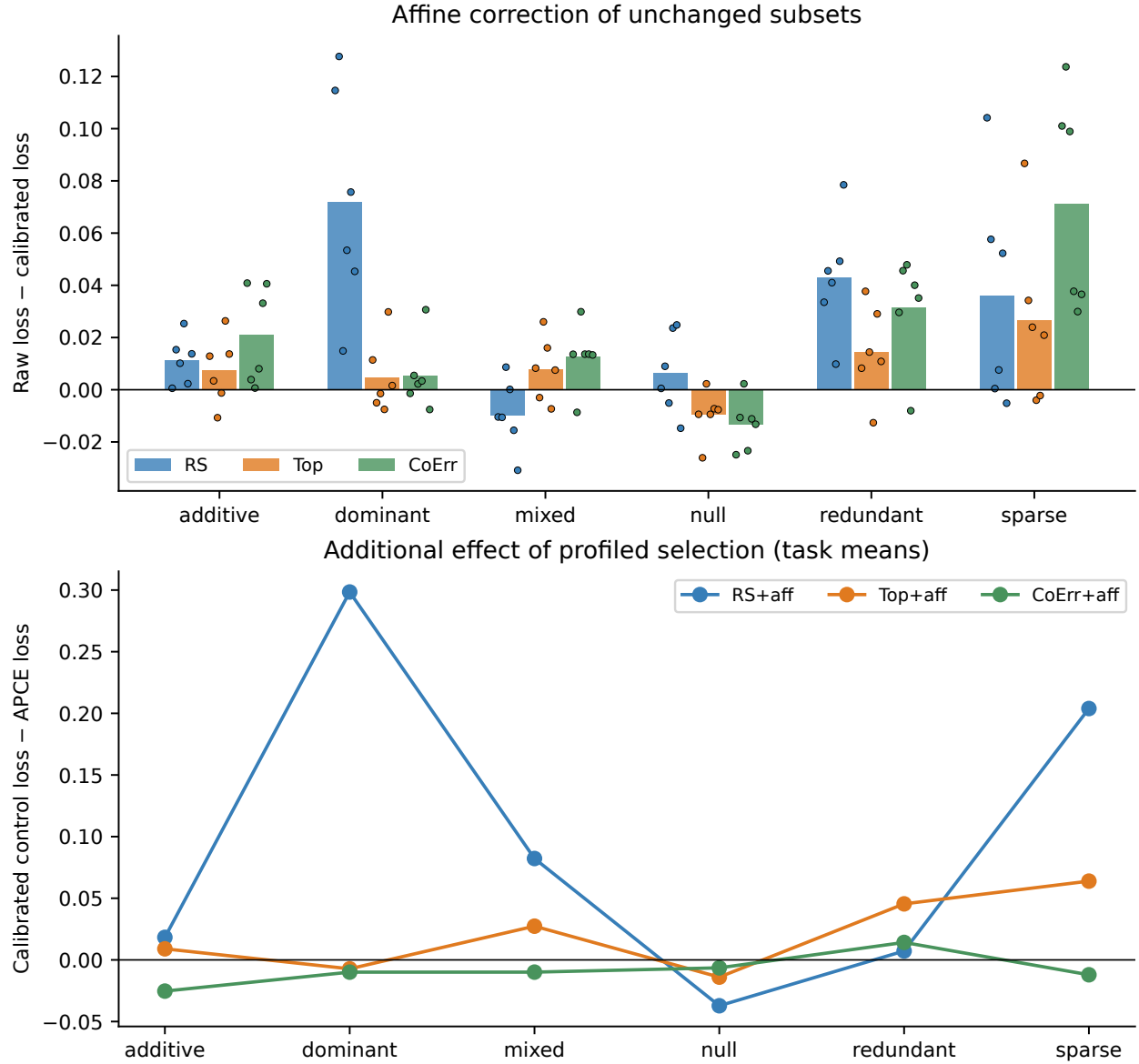


Figure 3: Affine diagnostic. Top: paired loss changes for unchanged subsets; dots show individual outer split effects and bars their means, not confidence intervals. Bottom: profiled selection relative to calibrated controls in task means. Positive values favor the added step. The null and most profiled co-error comparisons supply counterexamples.

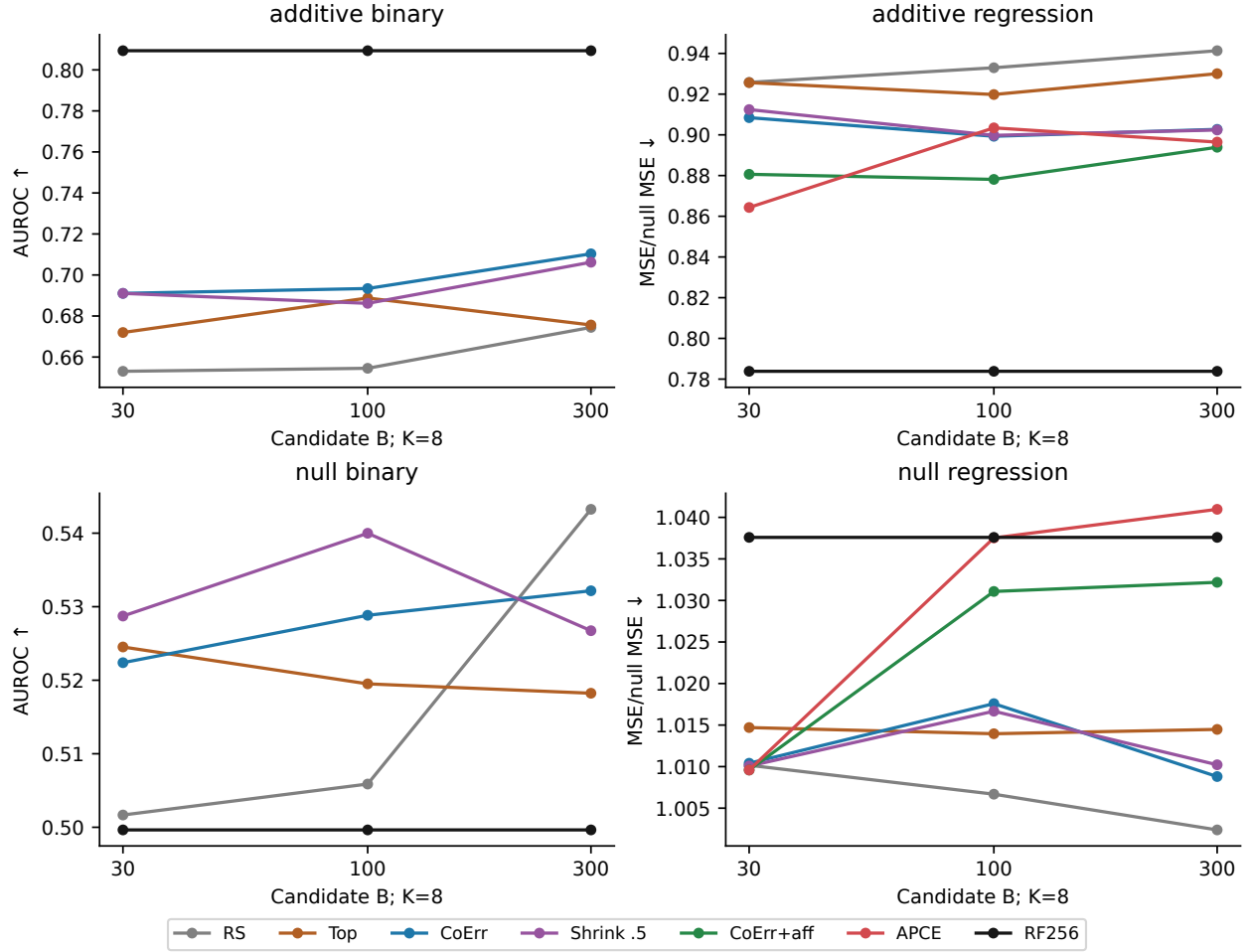


Figure 4: Candidate-pool scaling at fixed  $K = 8$ . Task means over six paired outer evaluations. Forest capacity is larger and budgets are unmatched; its unchanged predictions serve as a reference. Larger pools help binary additive co-error, but affine/profiled regression does not improve monotonically. Full split and generation-seed effects are stored with the artifacts.



mum squared-loss/co-error identity discrepancy was below  $10^{-12}$ . Direct squared-loss and co-error greedy validation objectives agree, confirming that their distinction supplies no additional mechanism.

## 8 Analysis and Next Discriminating Test

Additive regression illustrates the attenuation concern: normalized squared loss is 0.899 for unfiltered co-error, versus 0.784 for the unrestricted forest and 0.764 for CatBoost. This observation does not establish that attenuation is the sole cause. Experiment 002 confirms modest fixed-subset calibration gains while refuting a consistent advantage of profiled selection. Experiment 003 distinguishes improved validation fit from improved outer performance as search freedom grows. Raw OOF/test loss gaps combine estimation, training-size, and sampling effects, so this diagnostic alone cannot identify the source of optimism. The next test uses independent selection holdouts, comparing retained fitted functions with refitted specifications. Simultaneous binary AUROC certification for an independent fixed library is possible using a conservative bounded-differences bound [McDiarmid, 1989]; it supplies no automatic guarantee for OOF scores or subsequent refits. Linear baselines and oracle subspace controls are also needed to separate representation mismatch from selection failure on the known additive designs.

## 9 Limitations

This draft does not establish a novel algorithm or strong benchmark performance. OOF selection risks overfitting to a reused training prediction matrix. Refitting changes the fitted functions and may change their error dependence. IID row bootstrap screens do not justify guarantees under overlapping training folds, large candidate libraries, grouped observations, or temporal shifts. Sparse shallow trees can fail on interactions that need joint feature access. Only two generation seeds and one dimensional/sample-size setting have been tested in the completed studies. Static baselines lack tuning and resources are unmatched. Phase RSS includes the runner’s retained models and shared arrays, so it does not estimate isolated model memory. Synthetic regimes can share generation randomness; descriptive task resampling does not supply calibrated interval coverage. Linear/oracle controls and competitive tuned modern methods remain untested. Partial screening feasibility changes the support of comparisons; aggregate effects can be driven by one mechanism. These restrictions preclude real-world or broad algorithm-superiority claims.

## 10 Conclusion

The first development study does not establish a broadly competitive method. It reveals weaknesses in unadjusted bootstrap screening and in sparse-subspace aggregation, while co-error offers only modest and conditional gains over quality selection. Fixed-subset affine calibration helps some regimes, but profiled selection usually worsens test loss despite improving OOF fit. Increasing the candidate library has conditional benefits and can amplify this optimism at fixed retained size. Independent selection/refit controls remain necessary before any locked benchmark evaluation.

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